Summary of Fast Runs and Experiments

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Purpose

This note summarizes the fast Monte Carlo diagnostics for the IVQR project. All reported runs are **fast runs with $R = 10$ replications per design cell**. The main purpose is to diagnose estimator behavior, grid adequacy, and weak-identification problems before running full simulation.

The main estimators are:

Estimator	Role
Oracle IVQR	Benchmark estimator using oracle control information.
Post-selection IVQR	Feasible IVQR after Lasso-based control selection.
DML-IVQR	Orthogonalized/cross-fitted estimator for high-dimensional nuisance adjustment.

Baseline estimator runs

Baseline configuration: alpha grid $[-1, 3]$, 21 grid points, step size 0.2, $R = 10$.

Estimator	Coverage	CR length	Boundary hit	Disconnected CR	RMSE	MAE	Runtime
Oracle IVQR	86.0%	2.148	48.9%	43.7%	0.846	0.607	0.49s

Estimator	Coverage	CR length	Boundary hit	Disconnected CR	RMSE	MAE	Runtime
Post-selection IVQR	76.7%	1.849	46.9%	59.9%	0.893	0.647	13.68s
DML-IVQR	88.0%	2.642	61.5%	24.4%	0.986	0.753	27.39s

Main interpretation: 1. DML-IVQR gives the highest coverage, but partly because it produces wider confidence regions. 2. Oracle IVQR has better point accuracy. 3. Post-selection IVQR is the weakest estimator because it has the lowest coverage and the highest disconnected-region rate.

Coverage by DGP

DGP	Oracle IVQR	Post-selection IVQR	DML-IVQR
DGP1	90.4%	83.1%	87.9%
DGP2	81.0%	67.9%	89.0%
DGP3	86.7%	79.2%	87.1%

DGP2 is the main failure case. Post-selection IVQR performs especially poorly there.

Coverage by quantile

Quantile τ	Oracle IVQR	Post-selection IVQR	DML-IVQR
0.25	86.2%	77.5%	89.0%
0.50	89.0%	83.1%	87.7%
0.75	82.9%	69.6%	87.3%

The upper quantile $\tau = 0.75$ is the hardest case. This is expected because tail quantiles use less balanced information and are more sensitive to heteroskedasticity, extreme values, and weak effective identification.

Boundary hits by instrument strength

Instrument strength π	Oracle IVQR	Post-selection IVQR	DML-IVQR
0.10	97.2%	94.7%	98.3%
0.25	77.5%	69.4%	89.4%
0.50	19.2%	21.4%	51.7%
1.00	1.7%	2.2%	6.7%

Weak instruments create severe boundary problems. For $\pi = 0.10$, almost all confidence regions touch the grid boundary. Therefore, the baseline grid $[-1, 3]$ is too narrow for weak-IV designs.

Experiment A: wider alpha grid

Experiment A changed the grid to $[-2, 4]$ with 31 grid points and step size 0.2. The critical-value multiplier remained 1.0.

Estimator	Coverage	CR length	Boundary hit	Disconnected CR	RMSE
Oracle IVQR	86.0%	2.726	36.8%	50.2%	1.050
Post-selection IVQR	76.7%	2.337	36.0%	64.7%	1.092

The wider grid reduced boundary hits but did not improve coverage. Therefore, grid truncation is a real problem, but it is not the only problem. There is also weak-identification and post-selection instability.

Experiments B and C: alternative post-selection variants

Experiments B and C were run on a hard DGP2 subset. They are diagnostic only, not full-design experiments.

Variant	Coverage	CR length	Boundary hit	Discon-nected CR	Failed-alpha rate	Mean selected controls
B: aligned post-selection IVQR	55.0%	1.926	83.3%	93.3%	23.9%	63.2
C: post-selection quantile	68.3%	2.237	85.0%	95.0%	19.7%	47.5

Experiment B should be rejected. Experiment C is better than B, but still not strong enough to be a main estimator.

On the same DGP2 subset, the baseline methods performed as follows:

Method	Coverage	CR length	Selected controls
Oracle IVQR	78.3%	2.824	20.0
Baseline post-selection IVQR	75.0%	2.281	41.9
DML-IVQR	96.7%	3.590	–
Wide-grid post-selection IVQR	75.0%	3.229	41.9

DML-IVQR is conservative on the hard subset. Post-selection variants do not improve the baseline.

Experiment D: critical-value multiplier

Experiment D compared critical-value multipliers 1.1 and 1.2 on the baseline grid $[-1, 3]$.

Version	Oracle coverage	Post-selection coverage	Oracle CR length	Post-selection CR length
CV 1.10	87.0%	77.8%	2.217	1.909
CV 1.20	87.6%	78.6%	2.279	1.964

Multiplier 1.2 is slightly better than 1.1, but the improvement is small. It does not solve the undercoverage problem.

Post-selection diagnostic

Post-selection performance deteriorates when many controls are selected.

Selected controls	Rows	Coverage	CR length	Disconnected CR	RMSE
0–10	83	91.6%	1.893	31.3%	0.728
11–20	460	83.9%	2.091	45.9%	0.867
21–50	751	75.0%	1.790	66.2%	0.891
51+	146	54.8%	1.366	88.4%	1.051

This is one of the clearest results. When post-selection selects more than 50 controls, coverage collapses. This indicates that selection uncertainty is not adequately reflected in the confidence regions.

Main conclusions so far:

1. Post-selection IVQR is the least reliable estimator in the current implementation. Alternative post-selection variants in Experiments B and C do not solve the problem.
2. DML-IVQR gives the best coverage, but with wider regions and worse point-estimation accuracy.
3. Oracle IVQR remains the benchmark and has the best point accuracy, but it also undercovers.
4. DGP2 and $\tau = 0.75$ are the hardest cases.
5. Weak instruments generate frequent boundary hits, so the current alpha grid is too narrow.
6. Disconnected confidence regions are common and should be reported as a weak-identification diagnostic.